

CIPHER • v1.4 SECURE

CIPHER

Investigate syndicates with explainable intelligence

AI-Powered Criminal Network Intelligence Platform

Complete Pipeline, Screen Flow & Implementation Guide

PS 26189 — AI-Powered Criminal Network Analysis System

Tekathon 5.0 — 2026 | Theme: Blockchain & Cybersecurity | Team: Vedantra

From raw data source to courtroom-ready evidence: the full pipeline, every screen, every diagram

This document expands the CIPHER concept deck into a complete, buildable pipeline specification: how data moves from seven raw source types through NLP extraction, human-in-the-loop verification, blockchain-anchored evidence integrity, and graph analytics, into an interactive investigator dashboard — with a full screen-by-screen navigation flow, role-based access model, and phased implementation plan.

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1 Concept Overview

1.1 The gap CIPHER closes

- **Fragmented data:** FIRs, CDRs, financial records, vehicle records, and intelligence reports exist across disconnected sources, in disconnected formats.
- **Hidden relationships:** investigators must manually connect people, phone numbers, locations, vehicles, and organisations — a process that does not scale past a handful of entities.
- **Slow analysis:** large volumes of unstructured information make manual investigation time-consuming and prone to missed links, especially across cases that are never cross-referenced.

1.2 What CIPHER delivers

- **Key actors:** ranks influential nodes within a criminal network so an investigator knows who to prioritise, not just who exists in the data.
- **Hidden links:** surfaces indirect relationships across sources that are practically impossible to identify by manually cross-referencing documents.
- **Network patterns:** highlights clusters, unusual structures, and suspicious relationship patterns using graph analytics, not guesswork.
- **Investigative context:** lets an investigator move fluidly from one individual or one case out to the surrounding network, and back again.

1.3 Why CIPHER stands out

Principle	What it means in the build
Graph-first, not record-first	Traditional analysis looks at individual records in isolation. CIPHER's data model treats every entity and relationship as a first-class graph object from the moment it is extracted, so the graph is never a downstream visualisation bolted onto a records database — it <i>is</i> the database.
Multi-source intelligence	Instead of analysing FIRs, CDRs, financial records, and location data independently, CIPHER connects evidence across all seven source types into one unified network per case.
AI + graph analytics	AI handles extraction and entity resolution; graph analytics answers the actual investigative questions — who connects whom, how they connect, who matters most, and what patterns emerge — using deterministic, explainable algorithms rather than a black-box score.
Investigator-centric	CIPHER is not another raw-data dashboard. It turns complex, multi-source relationships into something an investigator can visually explore, query, and reason about, with every claim traceable back to its source evidence.

2 End-to-End Pipeline — Data Source to Investigator Insight

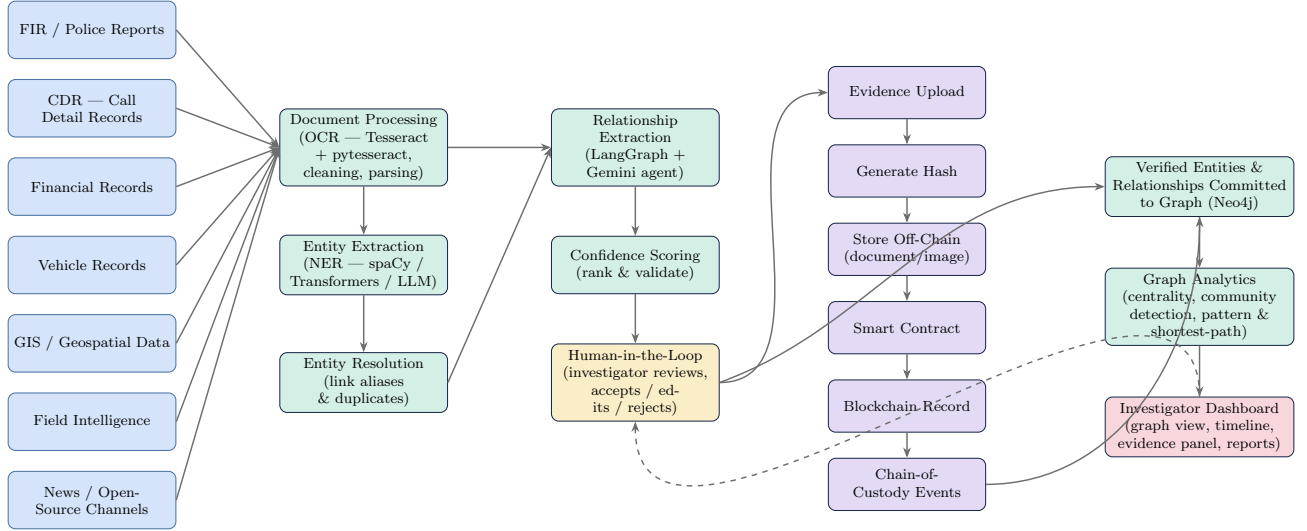


Figure 1: The complete CIPHER pipeline: seven raw data source types flow through document processing and NLP-based extraction, are scored for confidence, reviewed by a human investigator, anchored into an immutable blockchain evidence trail, and committed as verified graph objects that power the analytics and investigator dashboard. The dashed line shows the investigator’s ability to pivot back into review from any point in the graph.

2.1 Full pipeline diagram

2.2 Stage-by-stage notes

- **Document processing:** a router decides whether OCR is needed (Tesseract via `pytesseract` for scanned FIRs/reports), then applies text cleaning and structural parsing so downstream NLP receives clean, normalised text.
- **Entity extraction (NER):** spaCy or a transformer model tags person, location, organisation, and phone-number spans; structured identifiers (phone numbers, vehicle plates, account numbers) also get regex-rule reinforcement, since NER models handle these poorly on their own.
- **Entity resolution:** links aliases and near-duplicate mentions of the same real-world entity across different source documents (e.g. the same phone number appearing in a CDR and a financial record under two spelling variants of a name) before anything is scored.
- **Relationship extraction:** a LangGraph-orchestrated Gemini agent is prompted specifically to extract `(entity_A, relationship_type, entity_B, evidence_sentence)` tuples, constrained to only the entities NER already found — this prevents the model from inventing ungrounded relationships.
- **Confidence scoring:** every candidate entity match and relationship is stored with its source document, extraction method, and a numeric score *before* it ever reaches a human reviewer.
- **Human-in-the-loop:** the investigator accepts, edits, or rejects every AI-suggested entity and relationship. Nothing is treated as verified fact until this step completes — this is what keeps CIPHER explainable rather than a black box.

- **Blockchain evidence layer:** once accepted, the underlying evidence (document/image) is hashed, the original file is stored off-chain (hashes, not raw evidence, go on-chain), a smart contract registers the hash and metadata, and every subsequent access or edit is appended as a chain-of-custody event — giving investigators a tamper-evident audit trail suitable for legal proceedings.
- **Graph commit & analytics:** only verified entities/relationships are materialised as Neo4j graph objects; Neo4j Graph Data Science then computes centrality, Louvain community detection, pattern flags, and shortest paths on demand — these are computed, not AI-guessed, and are always labelled as such in the UI.
- **Investigator dashboard:** the graph view, timeline view, evidence panel, and report/export tools are how an investigator actually consumes all of the above — detailed screen-by-screen in Section 4.

3 Roles & Access Model

CIPHER’s landing page offers instant demo access via four named authorised roles, matching the department structure a real deployment would use:

Role	Demo account	What they can do
Inspector (Investigator)	<i>Inspector</i>	Owns case investigations: uploads evidence, explores the graph, builds timelines, requests entity review, and drafts reports for their own assigned cases.
Analyst	<i>Sanya</i>	Works across cases: cross-case entity search, pattern and anomaly review, contributes analytical notes, does not own case administration.
Supervisor (ACP)	<i>ACP</i>	Reviews and can override entity/relationship decisions in the human-in-the-loop queue, approves report exports, oversees their jurisdiction’s caseload and officer workload.
Admin	<i>Vikramaditya</i>	System administration: user and role management, DPDP/legal-compliance configuration, blockchain node/health monitoring, full system-wide audit trail, no direct case data access by default.

RBAC is enforced server-side on every request, scoped by (**role**, **action**, **case/jurisdiction**) — never inferred from the frontend. An Inspector never sees another jurisdiction’s cases; an Analyst can search across cases but cannot edit case administration; only a Supervisor or Admin can override a human-in-the-loop decision or approve a revocation/export at scale.

4 Screen-by-Screen Navigation Flow

4.1 Full navigation flowchart

4.2 Landing page — the front door

CIPHER’s public landing page (dark theme, CIPHER v1.4 SECURE badge in the top bar) is the first thing anyone sees, and it is deliberately built to demonstrate the product’s core claim before a single

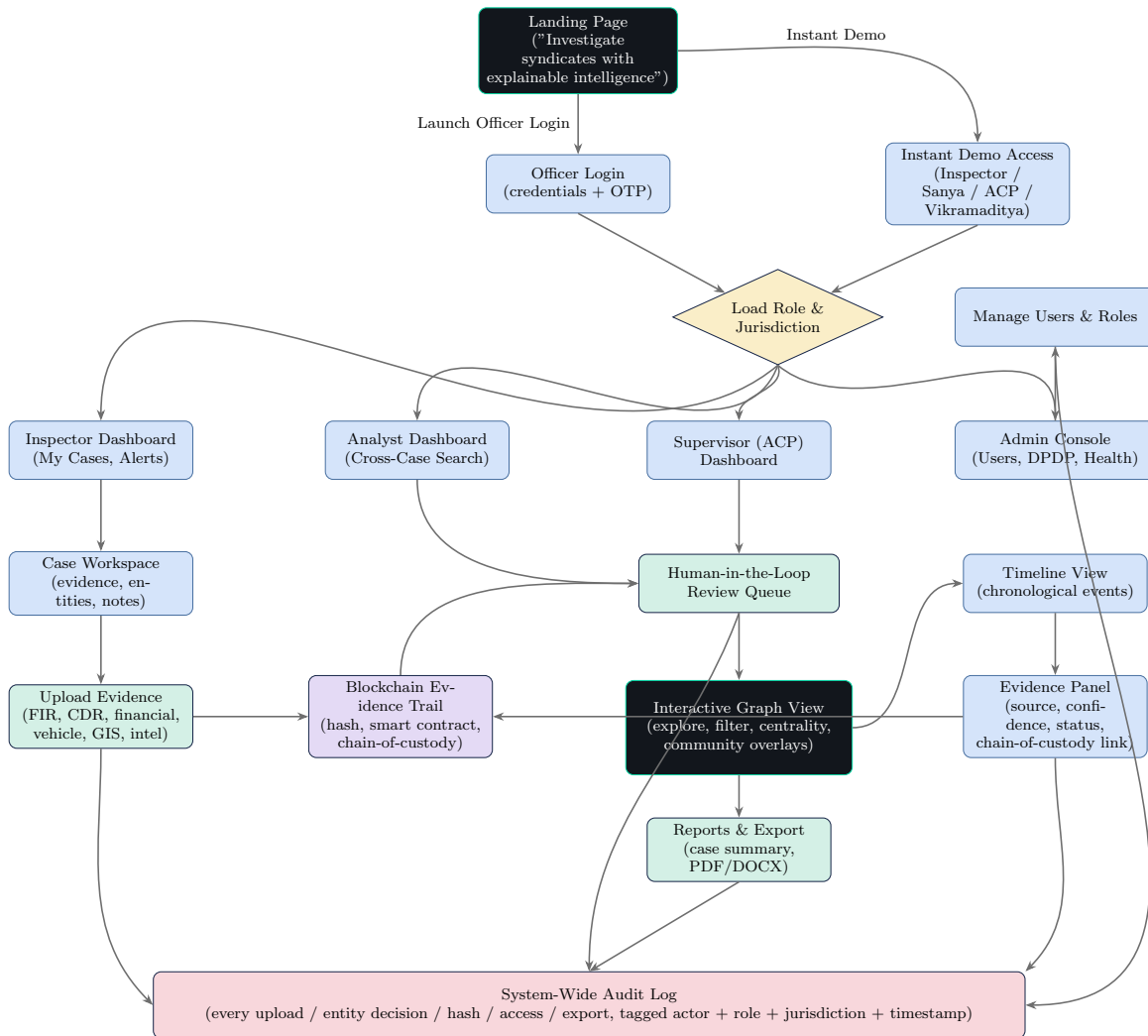


Figure 2: Full navigation flow: from the public landing page through role-based login (or instant demo access), into each role's dashboard, through evidence upload, blockchain anchoring, human-in-the-loop review, the interactive graph and timeline, evidence panel, report export, admin console, and the system-wide audit log that every action feeds.

click:

- Top navigation: *Capabilities*, *GIS Corridors*, *DPDP & Legal Dossiers*, *Department Roles* — each a real anchor into evidence of what the platform does, not marketing filler.
- Hero headline: *"Investigate syndicates with explainable intelligence"*, sub-headline explaining that the platform synthesises unstructured CDR phone records, hawala banking trails, tower-sector dumps, and field observations into interactive criminal network graphs and geospatial transit corridors.
- Two primary calls to action: **Launch Officer Login** (credentialed path) and **Explore Capabilities** (scroll-through product tour for non-authenticated evaluators/judges).
- A visible, honest secondary path: *"OR INSTANT ACCESS VIA AUTHORIZED DEMO ROLES"* — Inspector, Sanya (analyst), ACP (supervisor), Vikramaditya (admin) — so a judge or evaluator can experience every role without a real credential exchange, while still routing through the same role-aware backend logic a production login would use.

5 Blockchain Evidence Layer — Detailed Workflow

Why hash-anchoring, not full on-chain storage: storing raw evidence files on-chain is both unnecessary and impractical at scale. What a court or a supervisor actually needs is proof that a specific file has not changed since the moment it was entered into evidence — a hash comparison provides exactly that, at a fraction of the cost and complexity of storing the file itself on-chain.

6 Graph Analytics — What the Investigator Can Actually Do

Capability	What it surfaces
Network analysis (centrality)	Degree and betweenness centrality rank which entities are most connected or sit on the most paths between others — i.e. who the key actors and key intermediaries are.
Community detection (Louvain)	Automatically clusters the graph into sub-groups, surfacing syndicate structure that is not obvious from any single document.
Pattern detection	Flags structurally unusual relationship patterns (e.g. an entity bridging two otherwise-unconnected clusters) for investigator attention — always shown as a flag to review, never as an automatic accusation.
Shortest path	Given any two entities, computes the shortest connecting path through the graph — the direct, explainable answer to "how are these two people connected?"

All four are computed with Neo4j Graph Data Science on the verified graph only — never on unreviewed, AI-suggested data — and every result is presented as a labelled, computed overlay on the graph view, not as an unexplained score.

7 Technology Stack

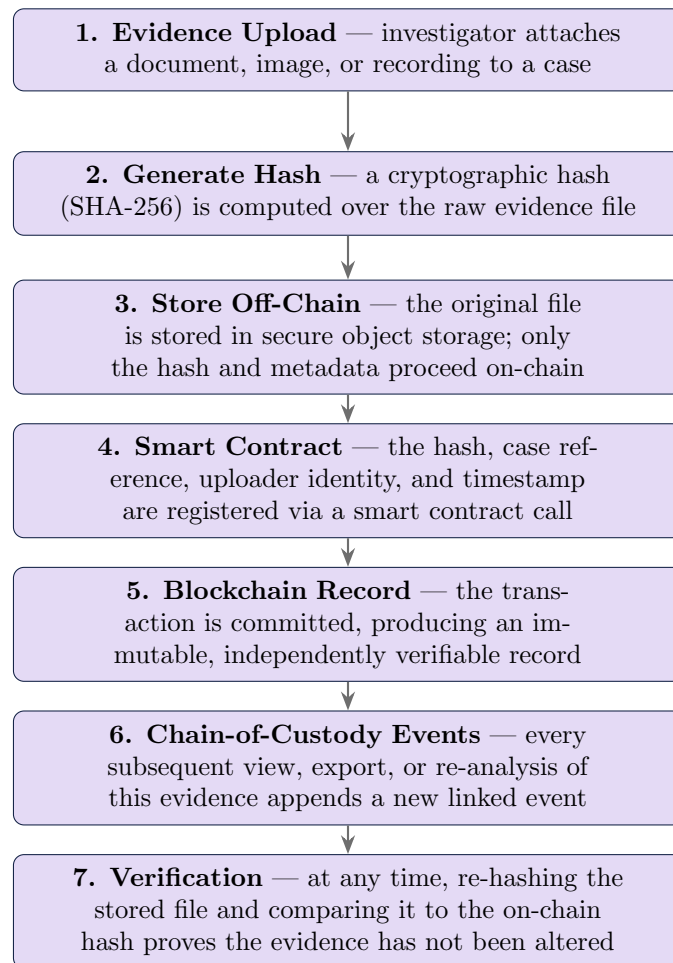


Figure 3: The blockchain evidence workflow. Only cryptographic hashes and metadata are ever written on-chain — raw evidence stays in conventional secure storage — so the chain stays lightweight while still making tampering mathematically detectable at any later point, including in a courtroom setting.

Layer	Technology	Why
Frontend	React.js + Cytoscape.js / react-force-graph	Component-driven UI with mature, interactive graph-rendering libraries supporting click events, filtering, and styling by attribute (confidence, verification state) out of the box
Backend / API	Python + FastAPI	Async-native, auto-generated OpenAPI docs, native WebSocket support for live pipeline progress
Real-time updates	WebSockets	Streams live status as documents move through OCR/NER/blockchain anchoring, without polling
Relational database	PostgreSQL	Users, cases, documents, audit log — all fundamentally transactional data needing ACID guarantees
Graph database	Neo4j	Native multi-hop traversal, built-in centrality/community-detection algorithms via Graph Data Science, Cypher query language
Cache / job queue	Redis	Decouples slow OCR/NLP/blockchain jobs from the request/response cycle
AI / NLP	Tesseract + pytesseract (OCR), spaCy / Transformers (NER), LangGraph + Gemini (relationship extraction)	Combines fast, well-understood OCR/NER with an LLM agent constrained to already-extracted entities, keeping the "agentic AI" claim honest and grounded
Graph analytics	Neo4j Graph Data Science (centrality, Louvain, shortest path)	Battle-tested implementations, avoids reinventing algorithms in application code
Blockchain evidence layer	Smart-contract hash anchoring (e.g. a permissioned or public EVM-compatible chain)	Lightweight, auditable, tamper-evident evidence trail without the cost of on-chain file storage
Security & access	JWT + RBAC	Standard, well-understood, jurisdiction- and role-scoped authorization checked server-side on every request
Deployment (demo)	Docker Compose	One-command reproducible setup for judges/mentors to run locally

8 Feasibility & Viability

Strengths	Limitations	Opportunities	Threats
AI-assisted entity extraction	Synthetic data	Authorised database integration	Privacy & data security
Graph-based analysis	AI outputs require validation	Cross-case network analysis	False entity matching
Human-in-the-loop verification	Data quality affects accuracy	Multilingual intelligence	AI-generated errors
Evidence-linked intelligence	—	Geospatial intelligence	Large-scale graph complexity

8.1 Risk mitigation

- **Confidence scoring** → reliable relationship ranking, so investigators know what to trust and what to verify first.
- **Human verification** → every entity/relationship is reviewed before it is committed to the graph.
- **RBAC + audit logs** → controlled data access, with every action traceable to an actor.
- **Docker + modular design** → scalable, reproducible deployment.

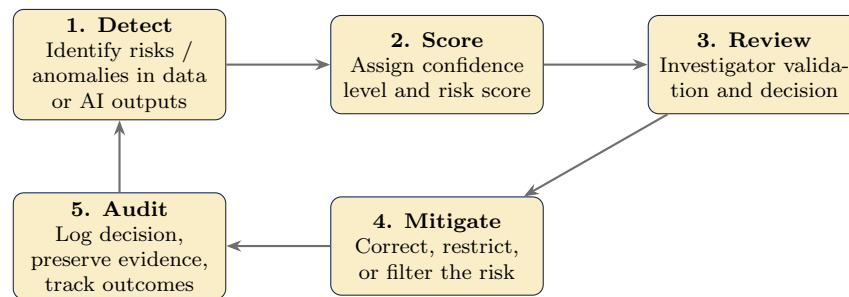


Figure 4: The risk-handling lifecycle: continuous monitoring and improvement, not a one-time check — every AI output is detected, scored, reviewed by a human, mitigated if needed, and logged, feeding back into the next detection cycle.

9 Impact & Benefits

Stakeholder	Impact	Benefit
Investigators	Faster relationship discovery, visual network understanding, evidence-linked analysis, reduced manual effort	Faster investigations: reduces manual relationship tracing
Police & intelligence units	Cross-case network analysis, identify key connectors, detect suspicious patterns, unified intelligence view	Hidden-link discovery: uncovers connections across entities

Stakeholder	Impact	Benefit
Social impact	Strengthens crime investigation, helps uncover hidden networks, supports evidence-based decisions, improves investigative coordination	Multi-source intelligence: unifies FIR, CDR, and intelligence data
Systemic impact	Multi-source intelligence fusion, scalable graph-based analysis, supports authorised data integration, creates an auditable analysis layer	Evidence-linked insights: supports verified, explainable decisions

The underlying application pattern, in every case, is the same four-entity composition: **Person + Phone** → **network nodes**, **FIR / Police Report** → **case anchor**, **CDR / Communication Network** → **relationship evidence**, **Vehicle + Location/Map** → **geospatial context** — CIPHER’s job is fusing all four into one explainable graph per case.

10 Step-by-Step Implementation Plan

1. **Phase 0 — Scaffolding.** Docker Compose (frontend, backend, Postgres, Neo4j, Redis), empty FastAPI app, empty React app, RBAC skeleton.
2. **Phase 1 — Auth & roles.** JWT login, OTP step, the four role-based dashboards (Inspector / Analyst / Supervisor / Admin) as empty shells, jurisdiction-scoped RBAC middleware.
3. **Phase 2 — Case & evidence intake.** Case creation, evidence upload UI, document-type router, storage adapter.
4. **Phase 3 — NLP pipeline.** OCR → NER → entity resolution → relationship extraction → confidence scoring, built and unit-tested as a pipeline independent of the web layer before wiring it to any UI.
5. **Phase 4 — Human-in-the-loop review queue.** The accept/edit/reject UI, writing verified decisions back to the same record the pipeline produced.
6. **Phase 5 — Blockchain evidence layer.** Hash generation, smart contract integration, chain-of-custody event logging, and the verification check — built and tested in isolation with a local/testnet chain before integrating into the upload flow.
7. **Phase 6 — Graph commit & analytics.** Only verified data reaches Neo4j; wire up centrality, Louvain community detection, and shortest-path via Neo4j GDS.
8. **Phase 7 — Investigator dashboard.** Interactive graph view, timeline view, evidence panel, reports & export.
9. **Phase 8 — Admin console & audit.** User/role management, DPDP & legal-dossier configuration, system-wide audit trail, blockchain/system health monitoring.
10. **Phase 9 — Synthetic data & polish.** Seed a realistic synthetic case universe (never real FIR/CDR/financial data), verify every screen against the design system, rehearse the demo script end-to-end.

11 Research & References

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5. Neo4j GraphAware — Intelligence Analysis
6. DataWalk — Law Enforcement Platform
7. ForceMetrics — Network Graph
8. "Explainable Graph Neural Network for AML," IJCNC, 2025
9. CCTNS Platforms
10. "Palantir AI and the Invisible Architecture of Impunity"
11. "AI/ML Fails to Combat Financial Crime: A Critical Review"

12 Summary Checklist Before Submission

- ☐ All four roles (Inspector, Analyst, Supervisor/ACP, Admin) can log in — via credentials or instant demo access — and land on the correct dashboard.
- ☐ Evidence uploaded to a case is hashed, stored off-chain, and produces a verifiable on-chain record with a visible chain-of-custody trail.
- ☐ Every AI-suggested entity/relationship is visibly distinct from a human-verified one, everywhere in the UI, and the review queue actually blocks unverified items from reaching the graph.
- ☐ The graph view supports centrality and Louvain community detection as labelled, togglable overlays — never presented as unexplained AI output.
- ☐ The timeline view filters the graph by date range.
- ☐ Every entity/relationship's evidence panel shows source document, extraction method, confidence, status, and a link to its chain-of-custody record.
- ☐ Re-hashing a stored evidence file and comparing it to the on-chain hash correctly detects tampering in a live demo.
- ☐ A case report can be exported as PDF/DOCX summarising the verified graph, timeline, and evidence.
- ☐ The system-wide audit log records every upload, entity decision, hash, access, and export, visible in full to Admin.
- ☐ You can state, in one sentence, why the platform hashes evidence rather than storing it on-chain.
- ☐ You can state, in one sentence, why entity resolution and human-in-the-loop review — not the blockchain layer alone — is what makes CIPHER's output trustworthy enough to act on.